

RAMESWARI MISHRA

Email: rameswarimishra1@gmail.com | +1 (573) 810-1335 | GitHub: [Rameswari Mishra. GitHub](#) | LinkedIn: [Rameswari Mishra. LinkedIn](#)

SUMMARY

Data Science Research Associate with six years of experience building scalable machine learning, informatics solutions, and data engineering systems across biomedical and enterprise environments. Expertise in predictive modeling, ETL pipelines, translational science technologies, SQL-driven analytics, and transforming large healthcare datasets into actionable insights. Open to relocation.

EDUCATION

Master of Science, Data Science and Analytics (Bio Health Computational Analytics), University of Missouri, Columbia, July 2024 (3.95/4)

Master of Technology, Computer Science and Engineering, NIST, Biju Patnaik University of Technology, INDIA, 2018(8.8/10)

Bachelor of Science, Computer Science and Engineering, RIT, Biju Patnaik University of Technology, Odisha, INDIA, 2006

TECHNICAL SKILLS

Programming & Databases: Python, R, Java, C/C++, JavaScript, PHP, SQL, MySQL, PostgreSQL, MSSQL, HTML5, XML, Web Services, Data Warehousing, Database Design

AI / Machine Learning: Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), Generative AI, Explainable AI (SHAP), Supervised & Unsupervised Learning

Data Analysis & Visualization: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn, Excel (Advanced), Tableau, Power BI, KPI Reporting, Dashboarding

Data Engineering & Systems: ETL/ELT Pipelines, Workflow Automation, EHR Systems, Data Integration, Biomedical Informatics, Project Management

Cloud & Engineering: AWS (S3, EC2), Azure, Docker, Databricks, Kubernetes, Data Streaming, CI/CD Pipelines, Snowflake, REST APIs, Model Deployment, Git, Version Control

Healthcare Data: Genomics whole-exome sequencing data, FASTA, VCF, BWA-MEM, Sam tools, GATK, EHR, ICD-9/10, REDCap, EDC, CTMS, eCRFs, DICOM, Longitudinal Dataset, Time Series Data, Triage, Survey Data, Billing Dataset, Claims Dataset, Psychiatric Data, Image Data

Data Quality & Governance: Data Validation, Data Lineage, Reproducibility, HIPAA Compliance

PROFESSIONAL EXPERIENCE

Data Science Research Associate & Instructor | University of Health Sciences and Pharmacy, St. Louis, MO | Aug 2024 – Present

Perioperative Depression Risk Prediction (4M+ Insurance Claims Dataset)

- Designed and built study-specific research datasets from 4M+ insurance claims; constructed pre- and post-surgery cohorts using data audits, data loads, and inclusion/exclusion criteria defined with clinical stakeholders.
- Validated all feature variables via systematic data integrity checks — profiling missingness, flagging ICD-9/10 discrepancies — and resolved issues with clinicians; maintained data definitions and process documentation throughout.
- Achieved 79% AUC (pre-surgery) and 68% AUC (post-surgery); applied SHAP explainability to surface antidepressant use, procedure types, and socioeconomic factors as key predictors.
- Applied t-SNE to cluster patient profiles by SHAP-derived predictor patterns, visualizing subgroup similarities to surface actionable pre- and intra-surgical mental health intervention targets
- Presented findings at APHA 2025 and contributed to ongoing grant writing and manuscript preparation

(<https://apha.confex.com/apha/2025/meetingapp.cgi/Paper/575298>)

Resting-State 120-Channel EEG Depression Prediction (MODMA Dataset)

- Developed clinically interpretable machine learning models for depression classification using resting-state 128-channel EEG recordings from clinically diagnosed patients and matched healthy controls
- Engineered linear, nonlinear, and statistical neurophysiological features, achieved 98% AUC and 92% accuracy, outperforming recent state-of-the-art research on the same dataset and demonstrating the potential of EEG-derived biomarkers for precision mental health diagnostics and clinical decision support

Instruction & Curriculum Leadership

- Instructor for Programming I & II (Python), Data Literacy, and Capstone Projects
- Designed curriculum covering OOP, data preprocessing, statistical analysis, ETL, and ML foundations
- Lead project management for undergraduate capstone teams — scoping research objectives, tracking milestones, and guiding teams from EDA through model deployment.
- Developed case studies and practical assignments to reinforce real-world data science workflows

Graduate Research Assistant | University of Missouri | Nov 2022 – July 2024

Project: Real-Time Fall Detection (Depth Video)

- Designed and deployed deep learning models (ConvNet + LSTM + YOLOv5) reducing false alarms by 11% in real-time fall detection systems.
- Optimized detection thresholds by integrating YOLOv5-based human detection with temporal LSTM modeling.

- Strengthened model robustness using InceptionV3 preprocessing, Docker containerization, and AWS S3-based data storage.

Data Science Intern | Johnson County Mental Health Center, Kansas City, KS | Jan 2024 – May 2024

- Executed ETL pipelines and supported deployment of ML models addressing Behavioral Health Crises, impacting 50+ high-risk patients monthly
- Automated SQL validation scripts using Python, reducing processing time from 3 hours to 30 minutes (20% productivity gain)
- Conducted EHR-based behavioral health analytics using ML algorithms and Tableau to inform intervention strategies

Data Science Intern | Blessing Health in Quincy, IL | May 2023 – Aug 2023

Predictive Model Pipelines Designing for Patient Panel Adjustment and creating dashboards:

- Built end-to-end predictive ML pipelines for patient panel adjustment using EHR data, including data cleaning, feature engineering, and hyperparameter tuning
- Achieved 0.72 test accuracy with XGBoost, outperforming baseline literature by 15%
- Partnered with MLOps to deploy the model into production
- Developed operational Tableau dashboards (workforce productivity, patient appointments, panel distribution) using PostgreSQL views, improving team efficiency by 20%

Automation Engineer | Inditek Pioneer Solutions, Hyderabad, India | Mar 2021- May 2022

- Automated large-scale data quality evaluation pipelines for fertilizer datasets, improving productivity by 15% and reducing processing time by 50%
- Designed and implemented CI/CD pipelines, automated reporting systems, increasing customer engagement and business value by 15%
- Coordinated cross-cultural, multilingual projects across geographically distributed teams, managing stakeholder communications and reporting to senior managers to deliver automated analytics solutions on time.

Software Engineer | Thrymr Software, Hyderabad, India | Nov 2018 - Mar 2021

- Developed enterprise web applications and analytics dashboards using Java, JavaScript, HTML5, XML, and PHP integrated with Oracle and SQL databases.
- Built ETL pipelines and REST-based web services to support business analytics and reporting solutions across multiple agile projects, improving productivity by 9%

SELECTED PROJECTS

Clinical NLP and LLM Pipeline for Structured Extraction from Hospital Discharge Summaries | [GitHub](#)

- Extracted structured attributes (CSV/JSON) from unstructured hospital discharge PDFs using rule-based NLP + LLMs; built interactive Power BI dashboard for clinical analytics with CI/CD-enabled workflow reproducibility.

Genomics Analysis for Lung Cancer Detection | [GitHub](#)

- Applied translational science tools (BWA-MEM, SAMtools, GATK) for whole-exome sequencing and genomic mapping, bridging basic genomic research with patient-centered clinical applications.
- Executed variant calling processes to identify potential biomarkers, explored biological pathway impacts to enhance understanding and facilitate accurate biomarker discovery.

Lung Cancer Prediction | [GitHub](#)

- Built a decision tree-based lung cancer prediction model with 0.94 accuracy using the Kaggle Survey Lung Cancer data.
- Conducted feature importance analysis to identify key predictors.

Adverse Events Analysis in Clinical Trial:

- Analyzed adverse events in a 62k-participant clinical trial using informatics tools (REDCap, EDC, CTMS, eCRFs), uncovering significant racial differences in event frequencies ($p < 0.05$).
- Enhanced clinical research analysis methods by identifying critical disparities, contributing to improved trial safety and efficacy.

Survival Analysis in Acute Myelogenous Leukemia Clinical Trial:

- Utilized Kaplan-Meier curves, log-rank test, and Cox Proportional Hazard model to conduct survival analysis on leukemia patients, demonstrating a significant treatment effect.
- Identified a hazard ratio of 2.50, indicating substantially prolonged remission in the chemotherapy group compared to controls ($\alpha = 0.10$), enhancing understanding of treatment efficacy.

Medical Image Classification for Tumor Identification:

- Applied Convolutional Neural Networks (CNNs) for the classification of MRI scans in DICOM format, focusing on accurate tumor identification.
- Attained a peak performance model with a loss of 0.2852 and an accuracy of 88.03%, achieved through precise hyperparameter adjustments.

CERTIFICATION:

WashU CAPS: Effective Business Presentations (Fall, 2025)

AWS Academy Graduate – Data Analytics ([AWS Academy Graduate - AWS Academy Data Analytics - Credly](#))

ISTQB Certified Tester – Foundation Level (2019)

Business Management Certification – (ICFAI, Tripura, India, 2009)